ARPIT TIWARI

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EDUCATION

IIT Bombay | B.Tech Mechanical + M.Tech AI & Data Science (Dual Degree), CPI 8.77 — 2019 – 2024

EXPERIENCE

Meesho — Bengaluru, India

Data Scientist 2 (Jul 2025 – Present) | Data Scientist 1 (Jul 2024 – Jun 2025) — *Jul 2024 – Present*

- **Leading the Address Enrichment & Prediction charter:** correcting and completing the addresses users have already written, and generating the full address through reverse geocoding when a user taps Locate me
  - **Address enrichment – pincode, city and state correction:** mentored the build of an **XGBoost**-based ranker that uses Spatial Brain features to fix these fields, raising correction accuracy from **77.9% to 95.5%** on 8.1M held-out orders and reducing return-to-origin (RTO) by **35 pp**
  - **Reverse geocoding (address prediction):** building an in-house, map-free engine to reconstruct a user's full address from GPS, aiming to write more deliverable addresses than the third-party provider MapmyIndia and so improve RTO and delivery metrics. The engine involves, in order: LLM + spatial deduplication of Spatial Brain entities; containment graph creation over their footprints to build the hierarchy tree (e.g. state → city → pincode → locality → sub-locality → landmark); landmark prediction and tree traversal to generate candidate addresses for a point; and ranking to serve the best one
- **Spatial Brain, Meesho's central geospatial knowledge base:** built it to discover geospatial entities and their footprints directly from hundreds of millions of delivery addresses, with no external maps; it powers address enrichment, reverse geocoding and the geocoder today, with many more use cases ahead
  - **Entity extraction:** built a span-detection model (**DeBERTa** encoder with classification heads, trained on LLM-generated teacher labels) that pulls localities, roads, shops and landmarks out of free-form address text
  - **Entity discovery:** aggregated delivery locations for each entity name, then used graph community detection (Louvain) to separate the different places where the same name occurs; for example, every HDFC Bank branch becomes its own entity
  - **Incremental maintenance:** built a daily incremental pipeline that updates the knowledge base by extending and merging existing entities and incorporating new ones, while keeping identifiers stable
- **GeoIndia, Meesho's in-house generative geocoder:** frames geocoding as a sequence-to-sequence problem, turning unstructured, multilingual Indian addresses into hierarchical H3 cells. Developed over five generations, each improving accuracy over the last:
  - **v1 – seq2seq geocoder:** fine-tuned FLAN-T5 and benchmarked QLoRA Llama-3-8B; deployed at Meesho scale, **outperforming Google Maps in all states**
  - **v2 – graph + language fusion** (CIKM '25): combined a graph transformer (Graphormer) over address-neighbourhood graphs built from field executives' last-mile delivery sequences with GeoIndia's FLAN-T5-based pre-trained language model, fused through a novel Key-Modulated Cross-Attention. Result: **+14%** relative accuracy within 200 m
  - **v3 SCOUTCG – personalised spatial RAG** (SIGSPATIAL '26): deployed geocoders are static and cannot adapt to new addresses without retraining. SCOUTCG makes the geocoder adaptive using each user's own location history: clustered **billions of in-app user location signals** into per-user spatial anchors (user–H3 graph with distributed Louvain community detection), and passed these anchors as spatial RAG context to the FLAN-T5-based geocoder alongside the address text in the prompt, served through a low-latency online feature store. Result: **+13%** relative accuracy within 200 m, with the largest gains on incomplete addresses
  - **v4 EPVA – entity-preserving tokenizer** (under review, EACL): general-purpose tokenizers split locality, village and landmark names into meaningless subwords. Designed a vocabulary adaptation that keeps each region's geographically informative entities as whole tokens while retraining the tokenizer on a national address corpus under a fixed vocabulary budget, then fine-tuned the LLM. Result: **+13.3%** relative accuracy within 200 m
  - **v5 ARGUS – generate-then-rank** (under review, AAAI): conceived and guided a generate-then-rank geocoder. Beam search over the v4 EPVA model generates candidate H3 cells, embedding-based retrieval (Qwen3) compares the address with historically successful deliveries in each cell, and confidence-aware fusion picks the final cell. Result: **+10%** / **+9%** relative accuracy at 500 m / 1 km over 12M deliveries; deployed pan-India
- **UAID, Meesho's address-to-location memory:** built the system that gives every address a unique ID and learns its true location from past deliveries, serving the location for existing users' previously delivered addresses

- **Address standardisation and deduplication:** normalised free-text Indian addresses and assigned each a unique, deterministic ID (UAID), so differently typed versions of the same address collapse into one record
- **Self-updating at scale: 300M+ UAIDs**, with **6M+** refreshed daily as new deliveries arrive
- **Confidence framework:** serves a location only when the address's past deliveries are tightly clustered

- **Unified Geospatial Clustering Framework** (AAAI '26): standard density clustering breaks when demand density varies widely across a region. Built an unsupervised clustering framework extending DBSCAN with Gaussian Mixture Models to choose clustering scales statistically and extract demand clusters from dense to sparse areas, on a distributed Spark pipeline validated across **200 last-mile dispatch centres**. It unlocked three logistics use cases:
  - **Self-pickup spot allocation:** powered Meesho's initiative to open pickup spots where users collect their own orders instead of home delivery; placed spots within 500 m of users, raising coverage from **20% to 95%**
  - **Density unbundling – field executive (FE) rate cards:** payouts that reflect regional delivery effort instead of flat per-order rates, cutting delivery cost by **3.2%** across 200 dispatch centres
  - **FE manpower allocation:** matched cluster-driven demand to staffing, showing **21%** of 200 centres understaffed (by ∼19%) and **79%** overstaffed (by ∼55%)
- **Strategic Hub Opening (Valmo, Meesho's logistics arm):** formulated network expansion as an operations-research **integer linear program** (facility location) that decides where to open new hubs and which areas each hub serves, minimising travel distance while prioritising high-RTO regions and unmet demand under capacity, service-distance and hub-budget limits; shipped it as an **interactive tool** used by ground ops to plan new hubs
- **Mentorship:** mentored **6** junior data scientists across Valmo and Geo Infra
  - **Network Design** (SIGSPATIAL '26): one data scientist on a multi-objective mixed-integer program (MILP) that plans vehicle routes and loading-dock slots together, warm-started from past plans, with a Claude-based LLM agent that turns depot managers' Hinglish instructions into model changes; deployed at all **80+ sort centres**: fleet −**20%**, cost per shipment −**14%**, same-day delivery **+17%**
  - **Competitive Bidding for First-Mile Procurement** (under review, AAAI): two data scientists on replacing the fixed rates negotiated with partner hubs for picking up sellers' orders with competitive bidding. Hubs bid the rate at which they will serve each area, and an optimisation model (MILP) assigns **50K+ sellers** to **250+ hubs** at lowest cost within capacity and service limits. Result: cost per shipment −**6.4%**, same-day pickup within ±2 pp
  - **Geo Infra:** two data scientists on GeoIndia geocoder models and one on Spatial Brain

INTERNSHIPS

Hilti Technology Solutions | *Software Development Intern* — ReactJS analytics dashboard — May 2022 – Jul 2022

Obbserv Online Services | *Data Analysis Intern* — KPI analytics (Tableau, Python) — May 2021 – Jul 2021

PUBLICATIONS

Published

- GeoIndia V2: A Unified Graph and Language Model for Context-Aware Geocoding, *CIKM '25* • Primary author
- A Unified Geospatial Clustering Framework to Identify Varying Density Clusters in E-Commerce Logistics, *AAAI '26* • Primary author
- SCOUTCG: Scalable Outline of User Trajectories for Contextual Geocoding, *SIGSPATIAL '26* • Primary author
- An Agentic Human-in-the-Loop Framework for Spatially-Aware Last-Mile Logistics Optimization, *SIGSPATIAL '26* • Mentor

Under Review

- EPVA: Entity-Preserving Vocabulary Adaptation for Generative Geocoding, *EACL* • Primary author
- ARGUS: Address Retrieval for Geocoding using Unified Semantics, *AAAI* • Mentor
- Optimization-Driven Competitive Bidding for E-Commerce First-Mile Procurement at Scale, *AAAI* • Mentor

SKILLS & ACHIEVEMENTS

Awards: Meesho Star Performer of the Quarter (twice), Star Team of the Quarter (twice)
Positions of Responsibility: Department Academic Mentor, IIT Bombay; Maintenance Secretary, Hostel 9

LLMs & NLP: LLM fine-tuning and pre-training (FLAN-T5, Llama-3, QLoRA), PyTorch, Hugging Face Transformers, tokenizer training, NER / span detection (DeBERTa, spaCy), embeddings, RAG / retrieval, LLM agents
Graph ML: graph transformers (Graphormer), cross-attention fusion, community detection (Louvain), graph construction
Unsupervised & Geospatial: clustering (DBSCAN, GMM), H3 spatial indexing, geocoding, reverse geocoding
Optimisation / OR: MILP, integer linear programming, facility location, vehicle routing, OR-Tools, CP-SAT, CPLEX
ML & Data: Python, SQL, PySpark, Databricks, scikit-learn, XGBoost, TensorFlow, Keras, online feature store